第三次作业（10月28日交、共8题）：课本正文第86页的题2，第88页的题5，第95页的13、14、15、16题，课本第107页的3、4题.

86-2

Suppose $E \subset \mathbb{R}$ is measurable, $a \in \mathbb{R}, \delta > 0$. If for all x such that $|x| < \delta$, we have $a + x \in E$ or $a - x \in E$, prove that $m(E) \geqslant \delta$.
Proof: For all $x \in B(0, \delta)$, we have $x \in (E - a) \cup (-E + a)$. Hence $B(0, \delta) \subset (E - a) \cup (-E + a)$, so

$$2\delta = m(B(0, \delta)) \leqslant m((E - a) \cup (-E + a)) \leqslant m(E - a) + m(-E + a) = 2m(E),$$

therefore $m(E) \geqslant \delta$.

88-5

Suppose $E \subset \mathbb{R}^n$. If for any $F \subset E \subset G$ where F is closed and G is open, we have

$$\sup_F m(F) < \inf_G m(G),$$

prove that E is not measurable.
Proof: See Exercise95-14.

95-13

Suppose $E \subset \mathbb{R}^n$, $E \subset H$ where H is measurable. If any measurable subset of $H \setminus E$ is null, must H be an equi-measure hull of E?
Proof: For any measurable G such that $E \subset G$, $G \cap H$ is measurable, so $N = H \setminus (G \cap H)$ is a measurable subset of $H \setminus E$ hence null. Then $m(H) = m(H \cap G) \leqslant m(G)$. Take G to be an equi-measure hull, then $m^*(E) \leqslant m(H) \leqslant m(G) = m^*(E)$ so H is also an equi-measure hull.

95-14

Prove that E is measurable iff for any $\varepsilon > 0$ there exists G_1, G_2 open such that $E \subset G_1, E^C \subset G_2$ and $m(G_1 \cap G_2) < \varepsilon$.
Proof: If such G_i exists, take G_n open and F_n closed such that $F_n \subset E \subset G_n$ and $m(G_n \setminus F_n) < 1/n$. Let $G = \bigcap G_n$ and $F = \bigcup F_n$ be Borel sets, then $F \subset E \subset G$ and $m(G \setminus F) = 0$. Hence $F \setminus E$ is null as a subset of the null set $G \setminus F$, so $E = F \cup (E \setminus F)$ is measurable.
Conversely if E is measurable, for any ε and n, take open set A_n such that $E_n = E \cap [-n, n]^k \subset tA_n$ and $m^*(E_n) + 2^{-n-1}\varepsilon > m(A_n)$, then $A = \bigcup A_n$ satisfy $m(A \setminus E) \leqslant \sum m(A_n \setminus E_n) < \varepsilon$. Analogously, we can find an open set B such that $E^C \subset B$ and $m(B \setminus E^C) < \varepsilon$. Therefore $m(A \cap B) < 2\varepsilon$.

95-15

Suppose $E \subset [0,1]$ is measurable, $m(E) \geqslant \varepsilon > 0$ with $x_i \in [0,1], i = 1, \cdots, n$, where $n > 2/\varepsilon$. Prove that there exists two points in E with distance equal to that of two points in $\{x_1, \cdots, x_n\}$.
Proof: Note that $A_i = E + x_i \subset [0,2]$, and if $A_i \cap A_j \neq \emptyset$, suppose $y_i + x_i = y_j + x_j$ where $y_i, y_j \in E$, then $y_i - y_j = x_j - x_i$, so the statement is valid. Note that A_i cannot be disjoint, otherwise

$$2 = m([0,2]) \geqslant m\left(\bigcup A_i\right) \geqslant nm(E) > 2$$

a contradiction.

95-16

Suppose $W \subset [0,1]$ is not measurable, prove that there exists $\varepsilon \in [0,1]$ such that for any measurable $E \subset [0,1]$ with $m(E) \geqslant \varepsilon$, $W \cap E$ is not measurable.
Proof: Otherwise for any n there exists $A_n \subset [0,1]$ measurable such that $m(A_n) \geqslant 1 - 1/n$ and $W \cap A_n$ is measurable. Let $A = \bigcup A_n$, then $m(A) \geqslant m(A_n) \geqslant 1 - \frac{1}{n}$ so $m(A) = 1$, and $W \cap A = \bigcup_k (W \cap A_k)$ is measurable. Hence $N = [0,1] \setminus A$ is null, and $W = (W \cap A) \cup (W \cap N)$, where $W \cap A$ is measurable, and so is $W \cap N$ as a subset of a null set. Therefore W is measurable, a contradiction.

107-3

Suppose $\{f_k(x)\}$ are measurable functions on E, then the converging points of f_k on E is measurable.
Proof: Let $B_{N,k} = \{x \in E : \forall n, m > N, |f_n(x) - f_m(x)| < 1/k\}$, then $B_{N,k}$ is measurable since $f_n - f_m$ is measurable so it is the intersection of measurable sets. Hence the converging points of f_k on E form the set $\bigcap_k \bigcup_N B_{N,k}$ which is measurable.

107-4

Suppose f is measurable on $E \subset \mathbb{R}$, G, F are open/closed sets in $\mathbb{R}$, then

$$E_1 = \{x \in E : f(x) \in G\}, E_2 = \{x \in E : f(x) \in F\}$$

are measurable.
Proof: By definition, the preimage of Borel sets are measurable. Or write open sets as the disjoint union of countable many intervals, and closed sets the complement of open sets.